General Filtering

General Filtering SOP

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Materials

- **Filtration System:** Located on the south wall of RH273, this consists of a vacuum pump, copper manifold, valves with yellow hands, and red tubing. The yellow handles control the openings (**parallel to copper pipe is off, perpendicular to pipe is on**).

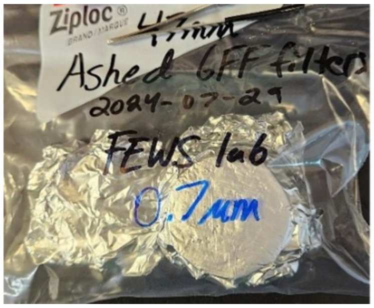
Figure 1: The two valves on the left are open.

- **Filtration apparatus:** including
  - the receiving container
  - filter disk
  - screw on funnel
  - (clear) stopper for the spout.

Important: the filter disk and funnel cup both have orange/red O-rings on the bottom, if they are missing the apparatus will have an inproper seal and leak.

- **GF/F ashed-filters (combusted):** these are wrapped in foil and located:

- SSC (suspended sediment concentration) needs pre-weighed and labeled filters, but they **DO NOT** need to be ashed.
- DOC (dissolved organic carbon) analysis (with or without SSC) needs ashed, labeled, and pre-weighed filters.



- **Blunt Tweezers:** In the cabinet above the filter station, or check nearby drawers if they aren't in the cabinet. Use the blunt tweezers to not tear filters. Get a separate pair for moving the dirty filters.
- **Kim wipes and DI water bottle:** used to wipe down funnel/tweezers.
- **Nitrile gloves**
- **Labeled bottles for the filtered sample**

Procedure

1. Set Up

1. Don gloves
2. **If doing SSC analysis:** weigh full sample bottles **WITHOUT** caps and record weight on lab data sheet.
3. **Before the first filtering:** Always pre-rinse a filter apparatus that were left in the cabinet (they accumulate dust). See below for rinsing procedure
4. Set up the filtration apparatus: set the filter disk on top of the receiving container. This should be placed with the rough side up and the o-ring contacting the catch bottle.
5. Using clean tweezers (rinse with DI water and wipe clean with a kim wipe) carefully remove a filter and carefully center it on top of the filter disk
6. Record the filter number in the filtering data table.
7. Screw the funnel piece onto the receiving container. Lightly press on the top while screwing the bottom corrugated plastic ring in place. Wipe any excess water on the funnel with a kimwipe. Make sure that the funnel is screwed on correctly and the O-rings are on the filter disk and funnel, or it will leak.

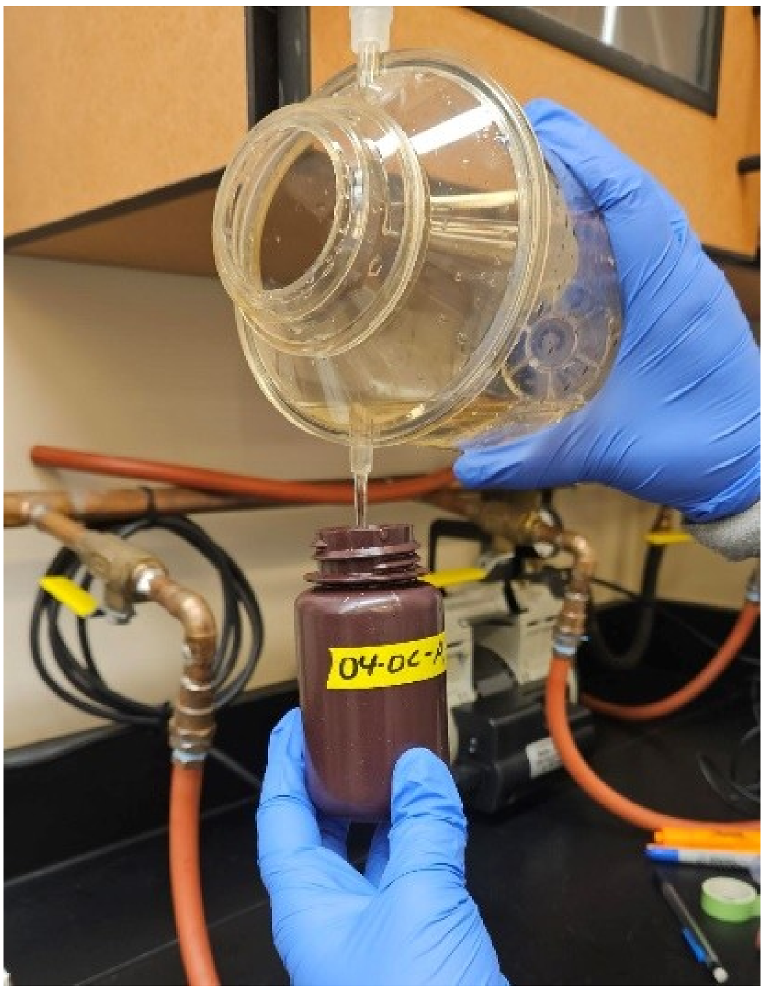
   > Tip: you can get the DI water during the sample rinse. Pour a little sample in with the pump off and swirl the container to catch the sides of the funnel

8. Connect the red tubing to an open spout on the catch bottle. Ensure the clear stopper is covering the other spout or it won't create a vacuum.
9. Ensure the apparatus is in a stable location (the red tubing can bend/move by itself and get knocked over)

10. Check that the yellow handles on the vacuum valves are pointed in the correct direction. **Parallel to the copper pipe is open and perpendicular is closed.** Close the valves that aren't being used or there will be no suction
11. Check if the vacuum trap (the clear chamber on attached to the inlet of the side of the vacuum pump) is empty. If it contains water, remove the trap and empty it before reattaching
12. Turn on the pump using the switch on the right side. Looking at the pressure gage on the pump, you want to aim for **25hg** and try not to exceed **30hg.**

2. Sample Rinse

1. Agitate the raw sample (**invert 5 times**) with the cap on before pouring. Aim for the center of the funnel.



2. Slowly pour until ~**50 ml** of sample has passed through the filter into the catch bottle (see image). **This is the sample rinse.**

3. Turn the vacuum pump off. Wait a few seconds for the pressure to stabilize. Remove red tubing from the apparatus. Unscrew the funnel and set it on a clean kimwipe. Remove the filter disk (keeping the filter undisturbed) by holding it by the edges. Place on kimwipe.
4. Over the sink gently swirl the receiving container to rinse the inside. Dump the water into the sink. The container is rinsed just **once.**

3. Sample Filtering

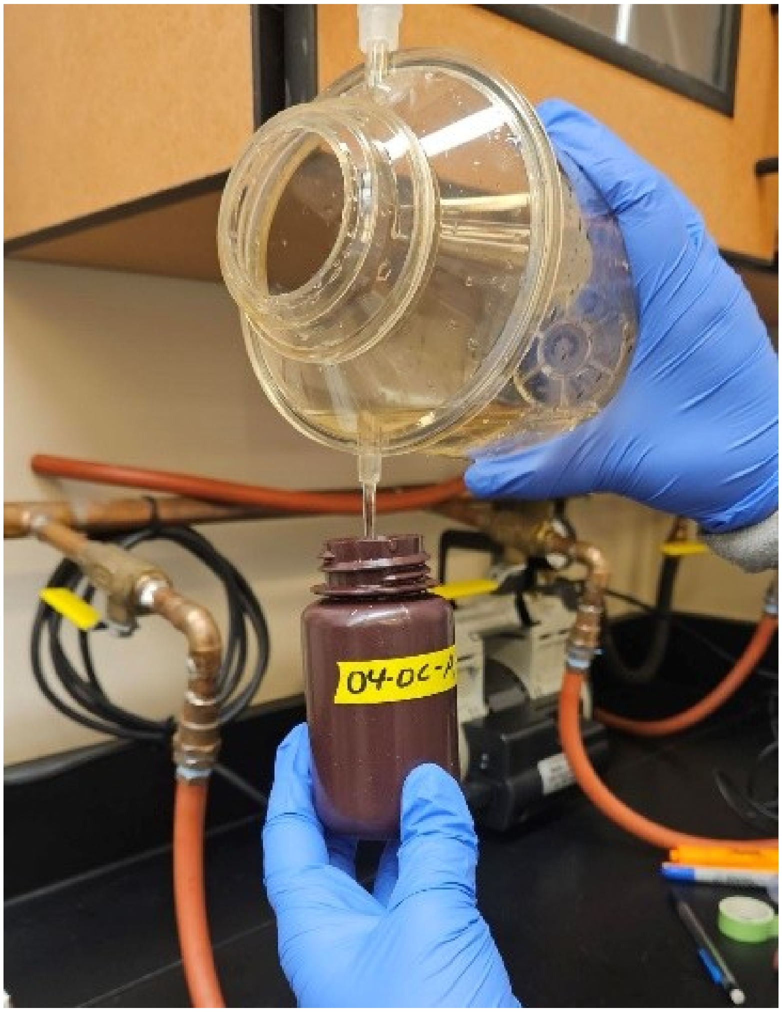
1. Reassemble the filtration apparatus again before connecting it to the red tubing.
2. Turn the vacuum pump on. Re-agitate your sample before slowly pouring until the sample runs out.
3. Once the sample is depleted, disassemble the apparatus.

4. Sample Distribution

1. Have your bottles ready for the filtered sample.

   See **A1. Common Bottle Types** table in the appendices for bottle types, how much to fill, and storage.

2. Make sure bottle numbers are recorded along with the filter date.
3. Pour about **5ml** of the filtered sample into its labeled bottle. You can use the spout or pour from the top.



4. Shake vigorously, dump in your bottle rinse in the sink. Repeat until it's rinsed **3 times.**
5. Pour the amount of sample filtrate needed for the bottle type (look at the table) and repeat rinsing and pouring for the other bottles.
6. Once all bottles are filled, any leftover sample filtrate can be dumped in to the sink.
7. Place the bottles into their corresponding coolers, when you are done for the day put the coolers in their correct spot (either fridge or freezer). If you aren't sure where they go, ask someone and leave them in the fridge for the time being.

5. SSC Analysis

1. If doing SSC analysis rinse the filter to get leftover sediment out of the bottle.

2. Make sure you have a tray ready covered in foil to place the filters on.
3. Reassemble the filter apparatus. Make sure you have finished collecting the sample filtrate.
4. Using a DI squirt bottle, squirt some DI water into the empty raw water sample bottle and agitate. Pour into the filter funnel. Repeat until the raw sample bottle has been rinsed 3 times.
5. Other method of just holding bottle over filter and using a squeeze bottle to flush the dirt out.
6. the rinsed filter on the tin foil covered tray. With a marker write the filter number next to it.

7. Place in drying oven at 105 degrees C for XX hours.

   Make sure the oven time has been scheduled on the Richardson Outlook calendar

8. The empty raw sample bottles that you just rinsed with DI water can be set aside to dry. **Do not** remove their labels. We usually hang them on the pegs above the sink. Be sure to leave a note

9. Once dry, weigh the bottles without the caps and enter their empty bottle weight value into your notes.

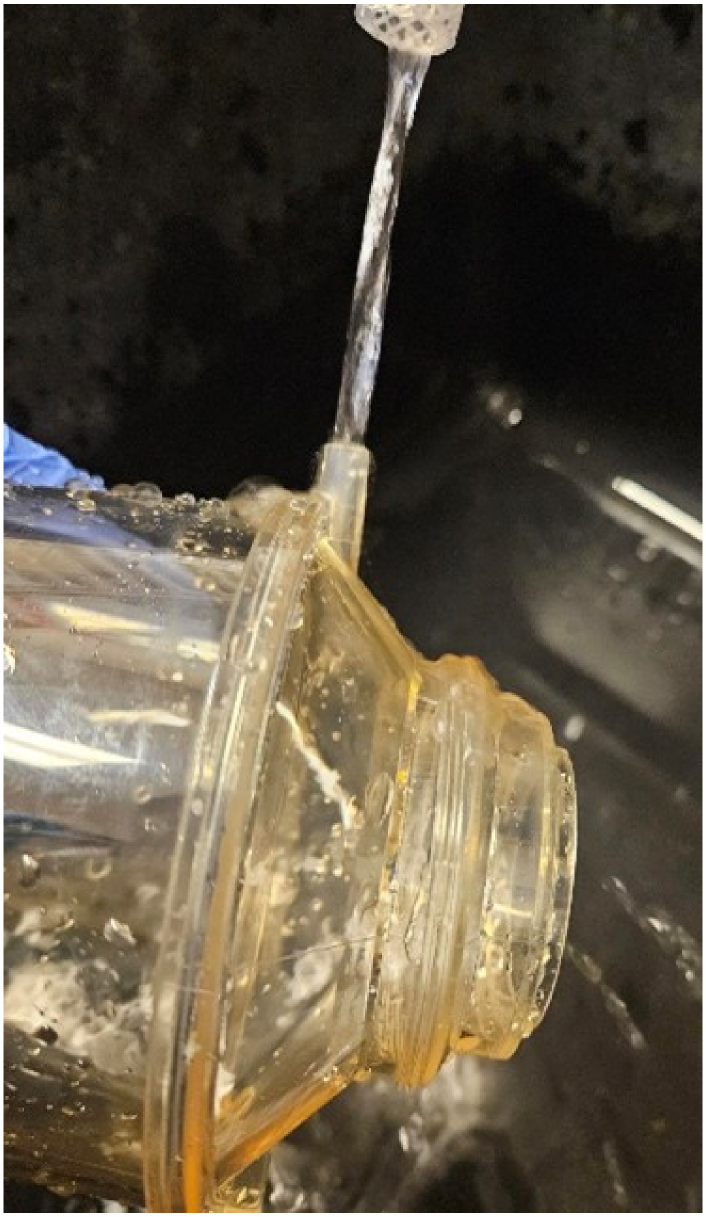
6. Rinsing

1. After each sample the apparatus needs to be rinsed

2. Keep gloves away from the inside of filter apparatus to avoid contamination. **Don't set the filter disk or funnel O-ring side down (part where it connects to the disk) in the sink. People use soap and dump dirt into the sink often.**

3. Using DI water rinse the receiving container **three times.** Be sure to rinse the spouts out.

   - You can rotate the container on its side to get it rinsed out



4. Rinse the funnel and filter disk off for about **10 seconds each** while rotating it.
5. Shake the excess water off and set aside to dry.

Appendices

A1. Common Bottle Types

Storm Sampling

Important: Frozen sample bottles will expand or break if overfilled!

Analysis	Bottle Type	How Much Sample	Storage
Aqualog (DOM)	30 mL or 125 mL	~30 mL	Fridge

Analysis	Bottle Type	How Much Sample	Storage
Shimadzu (TOC/TN)	125 mL	~80 mL (1 cm below neck). DO NOT OVERFILL.	Freezer
Levoglucosan	20 mL (glass)	~10 mL (1/2 full). DO NOT OVERFILL.	Freezer
CCAL (Nutrients)	125 mL	~80 mL (1 cm below neck). DO NOT OVERFILL.	Freezer
Excess - Drinking Water Treatment Testing (Node 3)	500 mL	Whatever is left over.	Fridge

A2. Post Run Clean Up

Important: Check with project lead before dumping any sample!

Analysis	Bottle Type	What to Do when Analysis is Complete
Aqualog (DOM)	30 mL or 125 mL	Dump once it's been confirmed there aren't any reruns.
Shimadzu (TOC/TN)	125 mL	